Architecture: today and next

One hazard. One Ultra call. The whole fleet knows.

Decision-making for robot fleets in warehouses and data centers: Nemotron 3 Ultra decides a new hazard once, fleet memory stores the decision by map cell, and every robot after that reuses it through Nemotron 3 Nano. People in the lane are never reused. Night Shift turns each day's decisions into tomorrow's reflexes.

ULTRA DECISION	GEMMA 4 PERCEPTION	SCHEMA-VALID	REPEAT SIGHTING
1.1–1.9 s	**0.7–1.2 s**	**100%**	**0 Ultra calls**
per call, live site	per frame, own H200 endpoint	Nano + Ultra decisions	settled by fleet memory

Measured 2026-10-08 on nemofleet.pages.dev, both scenarios. Robots, floors and camera frames are simulated; every model call is real.

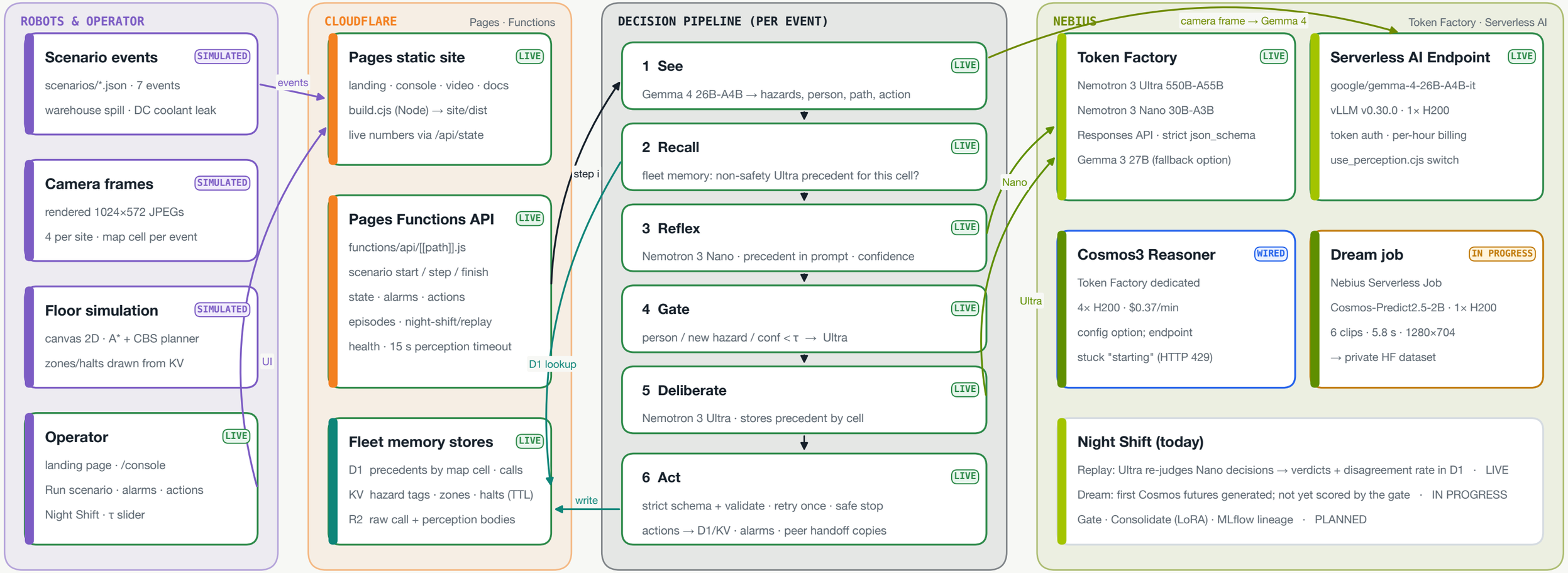
STATUS KEY LIVE WIRED IN PROGRESS PLANNED SIMULATED

CURRENT ARCHITECTURE · 2026-10-08

Today: what runs, end to end

LIVE WIRED IN PROGRESS PLANNED SIMULATED

Every box is in the repo or the live deployment. Arrows follow one scenario event from the robot to a fleet action.



TOOLING & OPS

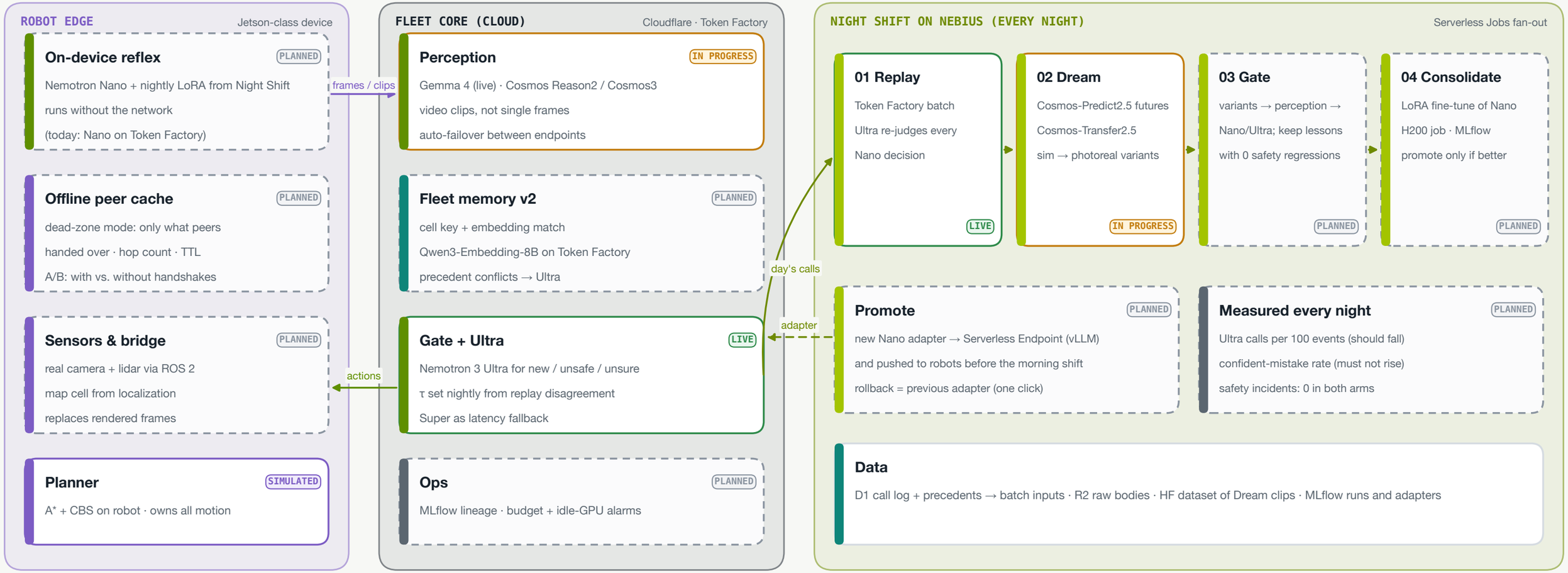
- ▸ Node 20+ (tools/*.cjs)
- ▸ wrangler (Pages · D1 · KV · R2)
- ▸ nebius CLI (endpoints · jobs)
- ▸ uv · Python 3.10/3.12
- ▸ Hugging Face (weights · dataset)
- ▸ puppeteer-core + Chrome (QA: 87 checks)
- ▸ preflight.cjs · use_perception.cjs
- ▸ GitHub (private until ready)

FUTURE STATE

Next: the fleet that learns overnight

LIVE WIRED IN PROGRESS PLANNED SIMULATED

Same decision core. The robot gets the reflex on-device, memory gets smarter, and Night Shift closes the loop on Nebius every night.



WHAT CHANGES

Today the fleet remembers within a day (precedents with a 24 h expiry). Next, Night Shift turns those decisions — plus thousands of Cosmos-imagined variants — into a better reflex model, so tomorrow Nano settles more on its own, Ultra is called less, and nothing about people is ever learned away.

COMPARISON

Today vs. next, component by component

COMPONENT	TODAY		NEXT	
Perception	Gemma 4 26B-A4B on own Serverless Endpoint (vLLM, H200); Cosmos3 wired	LIVE	Video clips; Cosmos Reason2 / Cosmos3 with auto-failover	IN PROGRESS
Reflex	Nemotron 3 Nano 30B-A3B on Token Factory, strict schema	LIVE	Nano + nightly LoRA on-device and on a vLLM endpoint	PLANNED
Deliberation	Nemotron 3 Ultra 550B-A55B; new / person / conf < τ	LIVE	Same, τ tuned nightly; Super as latency fallback	PLANNED
Fleet memory	Precedents keyed by map cell in D1, 24 h expiry, cross-run	LIVE	Cell + embedding match (Qwen3-Embedding-8B); conflict → Ultra	PLANNED
Peer handoff	Copies precedents with hop count in D1 + KV tags	LIVE	Offline dead-zone cache on the robot; measured A/B	PLANNED
Safety	Person → Ultra always; schema + retry + safe stop; never reused	LIVE	Unchanged; plus gate: zero safety regressions per adapter	PLANNED
Night Shift replay	Ultra re-judges Nano decisions, verdicts in D1	LIVE	Token Factory batch over the whole day	PLANNED
Night Shift dream	Cosmos-Predict2.5-2B job: 6 futures from camera frames	IN PROGRESS	Fan-out of hundreds of variants + Cosmos-Transfer2.5	PLANNED
Gate & consolidate	—	PLANNED	Score variants; LoRA fine-tune of Nano; MLflow; promote/rollback	PLANNED
Robots & sensors	Simulated robots, rendered frames, canvas floor	SIMULATED	ROS 2 bridge, real camera + lidar, Jetson-class edge	PLANNED
Operator UX	Console: runs, alarms, actions, memory KPIs; landing with live stats	LIVE	Clip playback, Dream verdicts, budget + idle-GPU alarms	IN PROGRESS
Ops & QA	QA suite (87 checks), preflight, perception switch, wrangler, nebius CLI	LIVE	MLflow lineage, nightly report, cost dashboard	PLANNED

INVENTORY

The whole tech stack

Everything NemoFleet uses today, grouped by layer. Status as of 2026-10-08.

NVIDIA models

Nemotron 3 Ultra 550B-A55B
deliberation · replay
LIVE

Nemotron 3 Nano 30B-A3B
reflex · memory reuse
LIVE

Nemotron 3 Super 120B-A12B
preflight · scenario gen
WIRED

Cosmos-Predict2.5-2B
Dream: futures
IN PROGRESS

Cosmos3 Super Reasoner
perception option
WIRED

Cosmos-Guardrail1
Cosmos safety filter
IN PROGRESS

Other models

Gemma 4 26B-A4B-it
perception (active)
LIVE

Gemma 3 27B-it
perception fallback
WIRED

Qwen3-Embedding-8B
memory v2 matching
PLANNED

Cosmos-Transfer2.5-2B
sim → real variants
PLANNED

Nebius

Token Factory
Responses API · strict schema
LIVE

Serverless AI Endpoint
Gemma 4 · vLLM · H200
LIVE

Serverless AI Job
Cosmos Dream · H200
IN PROGRESS

Dedicated endpoint
Cosmos3 · 4× H200
WIRED

Managed MLflow
adapter lineage
PLANNED

nebius CLI
endpoints · jobs · logs
LIVE

Cloudflare

Pages
landing · console · docs
LIVE

Pages Functions
API (JS, Workers runtime)
LIVE

D1 (SQLite)
precedents · calls · alarms
LIVE

KV
TTL hazard tags · zones
LIVE

R2
raw call + perception bodies
LIVE

wrangler
deploy · migrations · secrets
LIVE

Serving & ML runtime

vLLM v0.30.0
Gemma 4 OpenAI-compatible API
LIVE

CUDA 12.8 · cuDNN
Cosmos job image
LIVE

PyTorch · flash-attn
Cosmos inference
IN PROGRESS

uv · Python 3.10
Cosmos env
LIVE

Hugging Face Hub
weights · private dataset
LIVE

LoRA (PEFT)
Nano fine-tune
PLANNED

App & tooling

Node 20+ / CommonJS
tools · build · QA
LIVE

HTML · CSS · Canvas 2D
console + floor sim
LIVE

puppeteer-core + Chrome
QA: 87 checks
LIVE

marked
docs PDFs
LIVE

matplotlib
this PDF
LIVE

ffmpeg
explainer video · clip frames
LIVE